

Atomic Physics 2024

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Question 1

a i

Wavelength of light for transition between hydrogen states $n = 3$ and $n = 4$.

The wavelength of light needed to produce a transition of states in the Hydrogen atom is given by the Rydberg formula

$$\frac{1}{\lambda} = R_{\infty} \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad (1)$$

where $n_1 < n_2$ and $R_{\infty} = 1.1 \times 10^7$

In this case $n_1 = 3$ and $n_2 = 4$

$$\frac{1}{\lambda} = 1.1 \times 10^7 \left(\frac{1}{3^2} - \frac{1}{4^2} \right) \quad (2)$$

$$\frac{1}{\lambda} = 534722.2222 \implies \lambda = 1.87 \times 10^{-6} m \quad (3)$$

a ii

In the ground state, the electron is in the lowest possible energy state, and its wavefunction is confined to a small region around the nucleus (small volume). When the atom absorbs a photon of sufficient energy, the energy of the photon excites the electron to a higher energy state, meaning the electron is now in a different orbit or energy level. These higher-energy states correspond to larger volumes.

Uncertainty principle when the electron absorbs a photon its momentum increases and by the Uncertainty Principle the Uncertainty in position increases.

b i

We have the relation between the magnetic moment $\vec{\mu}$ the angular momenta $\vec{l}$ and $\vec{s}$

$$\vec{\mu} = -\frac{\mu_B}{\hbar} \left(\vec{l} + g_e \vec{s} \right) \quad (4)$$

where our constants of proportionality are: the Bohr magneton given by $\mu_B = \frac{e\hbar}{2m}$ and g_e is the g-factor ≈ 2 .

b ii

XXX

c

When considering electron configurations we need to keep in mind three rules

Pauli Exclusion Principle

no two electrons in an atom can have the same quantum numbers XXX. Each orbitals can have a maximum of **two** electrons with opposite spins.

Aufbau Principle

Electrons fill orbitals in order of energy levels

Hund's Rule

within a given subshell, electrons will occupy empty orbitals singly before pairing up.

The n th shell can hold up to $2(n)^2$ electrons.

$(2p)^1$ indicates there is 1 electron in the $2p$ subshell, which is possible as the max for $2p$ is 6.

If we examine the diagram below we see that clearly $(2p)^8$ doesn't make sense as the max for p is 6.

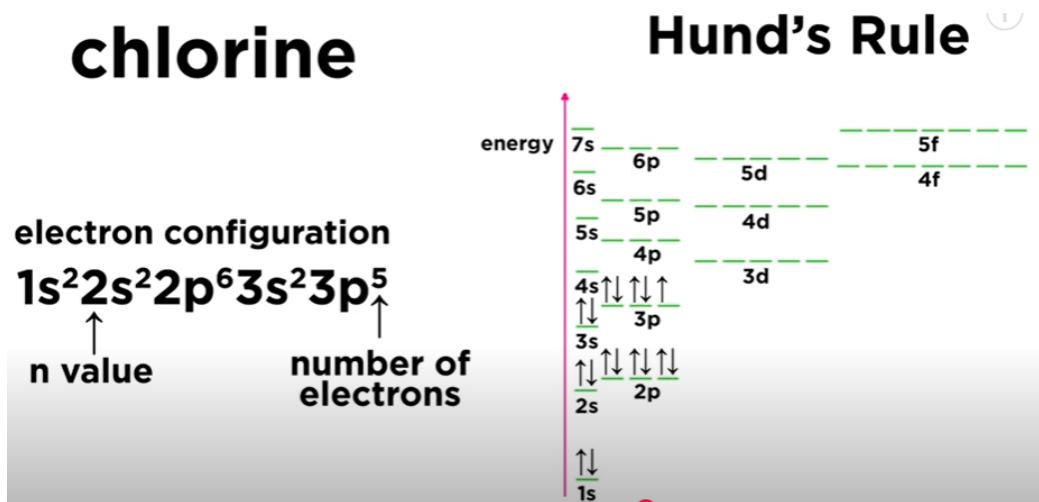


Figure 1: Here chlorine has 17 electrons. We fill from bottom to top. s has 1 bar (2 electrons) p has three bars (6 electrons), d has 5 bars (10 electrons). Each bar is filled by a single electron e.g up then once each bar at a give n is filled then we double up. We see the electron configuration has the letter to the power of the number of electrons filled.

c ii

In the $(2p)^2$ configuration. $l = 1$ $s = \frac{1}{2}$ 2 electrons (power).

To maximise the spin angular momentum $S = \sum_{i=1}^N s_i$ we know $S = s_1 + s_2$ or $S = |s_1 - s_2|$ (since we have 2 electrons) so either $s_1 = s_2 = \frac{1}{2}$ or $21 = s_2 = \frac{1}{2}$ Maximises S giving $S = 1$. The possible values of the total orbital angular momentum $\vec{L}$ is $L \in [l_1 + l_2, l_1 + l_2 - 1, \dots, |l_1 - l_2|]$. In our case for p orbital $l = 1$ so the possible values are 2, 1, 0. Thus the max $\vec{L}$ is $\vec{L} = 2$.

Question 2

a i

The $1s$ orbital is a dense cloud around the nucleus, while the $2s$ orbital is a shell within a shell i.e. a small dense region near the nucleus surrounded by a empty space (known as a node) then a diffuse outer cloud.

The $2s$ orbital is a higher state thus higher energy thus larger "volume". The $2s$ has a node of empty space between the peaks. In the node the prob of finding an electron is zero.

a ii

The reason why the solutions from Schrodinger's equation for the hydrogen atom differs slightly from the experimental value is because we assume that the proton always sits at $r = 0$ i.e. it doesn't move due to the influence of the electron. This is akin to assuming it has infinite mass.

To resolve this discrepancy we can replace the electron mass m_e with the reduced mass $\mu = \frac{m_e M}{m_e + M}$ where M is the

proton mass. Thus the electron will rather than orbit the proton it will orbit the COM of the system. One also could include relativistic treatments such as Dirac's equation.

b i

Given a state with $l = 2$ and $s = \frac{1}{2}$
The possible values of j is given by

$$j = |l - s|, |l - s| + 1, \dots, (l + s) \quad (5)$$

$$j = \frac{3}{2}, \dots, \frac{5}{2} \quad (6)$$

b ii

Showing

$$[J_z, \vec{s} \cdot \vec{l}] = 0 \quad (7)$$

$$[J_z, s_x l_x + s_y l_y + s_z l_z] = 0 \quad (8)$$

$$[J_z, s_x l_x] + [J_z, s_y l_y] + [J_z, s_z l_z] = 0 \quad (9)$$

We know that $J_z = L_z + s_z$ while L_z and s_z commute with each other so we only need to calculate the first 2 terms. Expanded these terms are:

$$[L_z, s_x l_x] + [s_z, s_x l_x] + [L_z, s_y l_y] + [s_z, s_y l_y] = 0 \quad (10)$$

Now we can use the known commutation relations.

$$[L_z, L_x] = i\hbar L_y \quad (11)$$

$$[L_z, L_y] = -i\hbar L_x \quad (12)$$

$$[s_z, s_x] = i\hbar s_y \quad (13)$$

$$[s_z, s_y] = -i\hbar s_x \quad (14)$$

Using the Jacobi identity we can split up the ones above into this and we see they vanish as required.

b iii

To estimate the expectation value of $\langle \vec{s} \cdot \vec{l} \rangle$ for hydrogen with $n = 2$, $l = 1$, $s = \frac{1}{2}$ and $J = \frac{3}{2}$.

We can find an expression for $\vec{s} \cdot \vec{l}$.

We know $\vec{J} = \vec{S} + \vec{L}$ so $\vec{J}^2 = (\vec{S} + \vec{L})^2 = \vec{S}^2 + \vec{L}^2 + 2\vec{S} \cdot \vec{L}$. Which gives us

$$\vec{S} \cdot \vec{L} = \frac{1}{2} [\vec{J}^2 - \vec{S}^2 - \vec{L}^2] \quad (15)$$

Thus now we have

$$\langle \frac{1}{2} [\vec{J}^2 - \vec{S}^2 - \vec{L}^2] \rangle = \frac{1}{2} [\langle \vec{J}^2 \rangle - \langle \vec{S}^2 \rangle - \langle \vec{L}^2 \rangle] \quad (16)$$

Which are the fundamental expectation values of QM

$$\langle \vec{J}^2 \rangle = J(J+1)\hbar^2 \quad (17)$$

$$\langle \vec{S}^2 \rangle = S(S+1)\hbar^2 \quad (18)$$

$$\langle \vec{L}^2 \rangle = L(L+1)\hbar^2 \quad (19)$$

Thus

$$\langle \vec{S} \cdot \vec{L} \rangle = \frac{1}{2} [J(J+1) - L(L+1) - S(S+1)]\hbar^2 \quad (20)$$

Plugging in our numbers we get

$$\langle \vec{S} \cdot \vec{L} \rangle = \frac{1}{2}\hbar^2 \quad (21)$$

There are in theory $\binom{6}{2} = 15$ as in the p subshell there are 6 available one electron states and each electron can be $\pm \frac{1}{2}$.

I will use the notation $^M L_J$. where M is the multiplicity $(2S + 1)$ i.e the number of J levels possible. These J levels have allowed states labeled by m_J . The number of states is given by $2L + 1$

$$^1 S_0 \quad (22)$$

$L = 0, S = 0$ one magnetic sublevel.

$$^3 P \quad (23)$$

$L = 1, S = 1$ and three possible J levels. Which are

$^3 P_0$ (w/ $m_J = 0$), $^3 P_1$ (w/ $m_J = -1, 0, 1$) $^3 P_2$ (w/ $m_J = -2, -1, 0, 1, 2$)

$$^1 D \quad (24)$$

$L = 2, S = 0$ Thus the compatible electronic states are 5 from $^1 D_2$ 1 from $^3 P_0$, 3 from $^3 P_1$, 5 from $^3 P_2$ and 1 from $^1 S$.

(c i)

We have that p corresponds with $l = 1$ so we have $m_l = -1, 0, 1$ but each particle can have spin of $\pm \frac{1}{2}$ thus we have 6 possible one-electron states. Pauli exclusion implies that we must.

(ii)